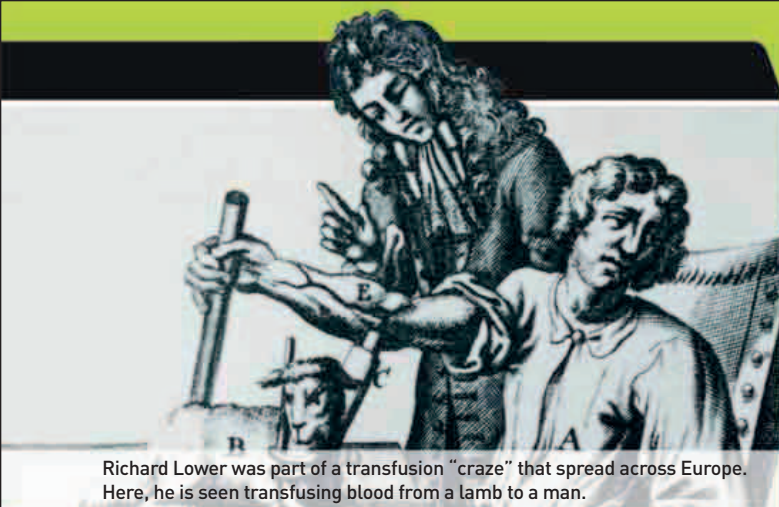




Richard Lower was part of a transfusion “craze” that spread across Europe. Here, he is seen transfusing blood from a lamb to a man.



sliding focus
composed
of leather

Replica of Newton's telescope

With its horizontal eyepiece, Newton's reflecting telescope was easier to use than traditional instruments. Its 1 1/4 in (3 cm) mirror reduced the optical imperfections seen in refracting telescopes.

movable
mount

until the work of Louis Pasteur in the 19th century on the introduction of organisms from the environment (see 1870–71).

In 1668, English chemist **John Mayow** (1640–79) developed a **combustion** theory countering earlier suggestions of burning occurring through the liberation of phlogiston. He saw that burning antimony—a metallic element—caused a gain, rather than a loss of weight. He suggested this came from a component of the air he called *spiritus igneo-aereus*. This idea anticipated the **discovery of oxygen** a century later.



A 19th-century engraving depicts Dutch naturalist Jan Swammerdam being plagued by a swarm of bees after removing their queen from the hive.



Joseph Wright's *The Alchemist*

This highly dramatic painting of 1771 is a fanciful recreation, rather than a true portrait, of Hennig Brand's accidental discovery of phosphorus.

mineralized remains of extinct organisms—found in these strata could be **sorted by age**.

The scientific study of insects arguably had its foundations in **Jan Swammerdam's** 1669 book *General History of Insects*. In this, the Dutch microscopist

GERMAN ALCHEMIST HENNIG BRAND

was searching for the “philosopher’s stone” that supposedly changed base metal into gold and, in 1669, he thought he had found it. But the glowing substance Brand discovered was **phosphorus**.

That year, Danish biologist and geologist **Nicolas Steno** (1638–86) explained that as sediment layers accumulated, old rock strata were overlain by newer ones. This meant that **fossils**—

described the larval and pupal stages of **insect life histories**.

In 1670, English chemist

Robert Boyle poured acid onto a metal and obtained inflammable air. Boyle had isolated **hydrogen**.

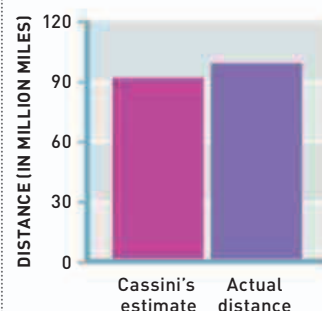
Cassini's work on **astronomical dimensions** in 1671 included his computation of the Earth–Mars distance, which gave the first indications of the Solar System's size. His 1672 calculation of the Earth–Sun distance is close to current estimates.

“IT IS UNWORTHY OF EXCELLENT MEN TO LOSE HOURS LIKE SLAVES IN THE LABOR OF CALCULATION.”

Gottfried Leibniz, German philosopher and mathematician, 1685

In 1672, **Isaac Newton** presented a paper to the Royal Society on his observations of the **rainbow of colors that make up white light** (see panel, opposite). He was subsequently elected a fellow of the Society, but Curator of Experiments Robert Hooke criticized Newton's paper, triggering an ongoing dispute between the two men.

In 1673, German mathematician **Gottfried Leibniz** created a **calculating machine** and presented it to the Royal Society. In the same year, Dutch astronomer **Christiaan Huygens**, inventor of the pendulum clock, published the **mathematical analysis of pendulum motion**, showing how length and weight affect swing.



Earth–Sun distance

Cassini's religious faith made him resist the idea of a Sun-centered Universe, but his views changed as he computed astronomical distances.

1668 English chemist **John Mayow** suggests air contains “aerial spirit”

1668 English mathematician **John Wallis** describes the law of **conservation of momentum**

1668 Italian physician **Francesco Redi** disproves theories of **spontaneous generation** of maggots

1668 **Newton** builds the **first optical reflecting telescope**

1669 Dutch microscopist **Jan Swammerdam** publishes *General History of Insects*

1669 German alchemist **Hennig Brand** discovers a new element, later identified as **phosphorus**

1669 **Steno** describes the **formation of rock strata** as a way of comparing the age of fossils

1670 English chemist **Robert Boyle** isolates **hydrogen**

August 1672 **Cassini** determines the **Earth–Sun distance**

1672 **Newton** publishes his first paper on **light**, describing the spectrum of colors

1673 Dutch astronomer **Christiaan Huygens** publishes *On the Pendulum Clock*

1673 German mathematician **Gottfried Leibniz** designs a **calculating machine**